

Chapter 2

Bank activities and services

Learning objectives

- To understand what modern banks do
- To describe the main services offered by banks
- To understand the importance of the payment system
- To understand the growing relevance of FinTech
- To identify the impact of FinTech on the banking industry

2.1 Introduction

This chapter offers some insights into the nature of the banking business. It reviews the main services offered by banks (loans and deposits, and payment services) as well as a wide range of additional services, such as insurance and investment services. In this context, special attention is given to the significant changes in payment systems with an extensive discussion on the major instruments and services that modern banks provide to their customers from plastic money to e-banking. The last section provides a brief introduction to FinTech and its growing importance for the banking industry.

2.2 What do banks do?

We have seen in Chapter 1 that financial intermediaries channel funds from units in surplus to units in deficit. In order to better understand how banks work, we need to examine their assets and liabilities. Table 2.1 summarises a typical retail bank balance sheet (details of banks' balance sheet and income structure are presented in Chapter 9).

For traditional retail banks, the main source of funding is customer deposits (reported on the liabilities side of the balance sheet); this funding is then invested in loans, other investments and fixed assets (such as buildings for the branch network), and it is reported on the

Table 2.1 A simplified bank balance sheet

Assets	Liabilities
Cash	Customer deposits
Liquid assets	Equity
Loans	
Other investments	
Fixed assets	
Total	Total

assets side of the balance sheet. The difference between total assets and total liabilities is the bank capital (equity). Put very simply, banks make profits by charging an interest rate on their loans that is higher than the one they pay to depositors. As with other companies, banks can raise funds by issuing bonds and equity (shares), and saving from past profits (retained earnings). However, the bulk of their money comes from deposits; it is this ability to collect deposits from the public that distinguishes banks from other financial institutions, as explained in Section 2.3.

2.3 Banks and other financial institutions

Banks are **deposit-taking institutions (DTIs)** and are also known as monetary financial institutions (MFIs). Monetary financial institutions play a major role in a country's economy as their deposit liabilities form a major part of a country's money supply and are therefore very relevant to governments and central banks for the transmission of monetary policy (see Chapter 5). Banks' deposits function as money; as a consequence, an expansion of bank deposits results in an increase in the stock of money circulating in an economy. All other things being equal, the money supply, that is the total amount of money in the economy, will increase.

The monetary function of bank deposits is often seen as one of the main reasons why DTIs are subjected to heavier regulation and supervision than their **non-deposit-taking institution (NDTI)** counterparts (such as insurance companies, pension funds, investment companies, finance houses and so on).

One further feature that distinguishes monetary financial institutions from other financial corporations lies in the nature of financial contracts: deposit holdings are said to be *discretionary*, in the sense that savers can make discretionary decisions concerning how much money to hold and for how long. Depositors are free to decide the frequency and amount of their transactions. Meanwhile, holding assets from other financial institutions requires a contract which specifies the amount and frequency of the flow of funds. For example, the monthly contributions to a pension fund or to an insurance provider are normally fixed and pre-determined. Therefore the flow of funds in and out of other financial intermediaries is described as *contractual*.

Figure 2.1 illustrates the classification of financial institutions in the UK. However, it is important to keep in mind that there is no unique, universally accepted classification of financial intermediaries. Furthermore, distinctions are becoming blurred as deregulation, financial conglomeration, advances in information technology and financial innovation, increased

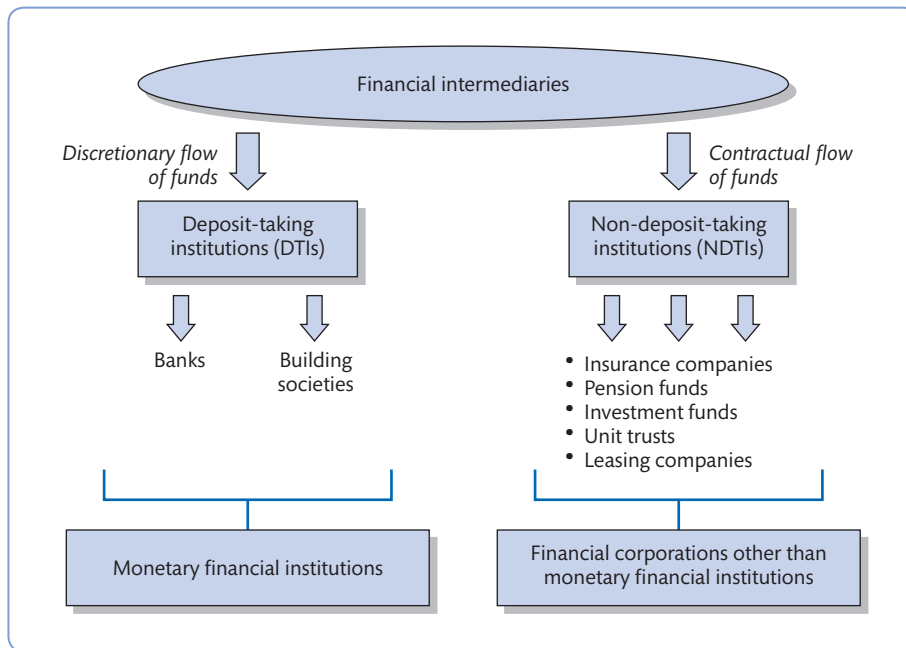


Figure 2.1 Classification of financial intermediaries in the UK

competition, globalisation and the fallout from the global financial crisis of 2007–2009, have all contributed to change the industry.

All countries have regulations that define what banking business is. For example, in all EU countries banks have been permitted to perform a broad array of financial services activity since the early 1990s, and since 1999 both US and Japanese banks are also allowed to operate as full-service financial firms. A good example of the breadth of financial activities that banks can undertake is given by the UK's Financial Services and Markets Act 2000, which defines the range of activities that banks can engage in, including:

- accepting deposits;
- issuing **e-money** (or digital money), i.e. electronic money used on the internet;
- implementing or carrying out contracts of insurance as principal;
- dealing in investments (as principal or agent);
- managing investments;
- advising on investments;
- safeguarding and administering investments;
- arranging deals in investments and arranging regulated mortgage activities;
- advising on regulated mortgage contracts;
- establishing and managing collective investment schemes (for example investment funds and mutual funds);
- establishing and managing pension schemes.

Conglomeration has become a major trend in financial markets, emerging as a leading strategy of banks. This process has been driven by technological progress, international

consolidation of markets and deregulation of geographical or product restrictions. In the EU, financial conglomeration was encouraged by the Second Banking Directive (1989), which allowed banks to operate as **universal banks**, enabling them to engage, directly or through subsidiaries, in other financial activities, such as financial instruments, factoring, leasing and investment banking. In the US, the passing of the Gramm–Leach–Bliley Act in 1999 removed the many restrictions imposed by the Glass–Steagall Act of 1933. Since 1999 US commercial banks can undertake a broad range of financial services, including investment banking and insurance activities. Similar reforms have taken place in Japan since 1999. As banks nowadays are diversified financial services firms, when we think about banks, we should now think more about the particular type of financial activity carried out by a specialist division of a large corporation rather than the activity of an individual firm.

In the immediate aftermath of the 2007–2009 financial crisis, a growing number of academics and policy makers debated on whether the size and permissible activities of financial institutions should be re-constricted due to increased systemic risk. Narrowing the scope of large financial institutions would require the re-introduction of firewalls like those imposed by the Glass–Steagall Act to limit the risks that depositors are exposed to. Post-crisis regulatory reforms in the UK, for example, have included the ‘ring-fencing’ of retail banking activities, following the recommendations of the Independent Commission on Banking (ICB), also known as the Vickers Commission (see Chapter 13).

BOX 2.1 UNIVERSAL BANKING: AN OPTIMAL MODEL OF BANKING?

A universal bank is an institution which combines its strictly commercial activities with operations in market segments traditionally covered by investment banks, securities houses and insurance firms, and this includes such business as portfolio management, brokerage of securities, underwriting, mergers and acquisitions. A universal bank undertakes the whole range of banking activities. Under the universal banking model, banking business is broadly defined to include all aspects of financial service activity – including securities operations, insurance, pensions, leasing and so on. Under this model, a bank will be able to serve the diverse needs of its customers and generate fee and commission income, which is not subject to the same kind of credit, maturity and other risks associated with income derived from lending and trading activities. Because of the easy access and convenience universal banks offer, they tend to attract customers along a very wide spectrum of financial sophistication and risk appetite. Some of these customers go to banks to conduct very basic banking transactions, such as cash withdrawal, remittances, applications for loans, credit cards, etc. However, as universal banks seek to expand fee and commission income in distributing investment products, there may be a risk that some banks or their frontline staff could become overly keen to market products to customers without sufficient regard to the customers’ own circumstances.

One example of this type of behaviour is the PPI (payment protection insurance) mis-selling in the UK. PPI was designed to cover repayments in certain circumstances, for example when people were made redundant or couldn’t work due to an accident, illness, disability or death. However, PPI was often mis-sold: the Financial Conduct Authority (FCA) reports that as many as 64 million PPI policies have been sold in the UK, mostly between 1990 and 2010, some as far back as the 1970s. Critics say the banking industry began aggressively selling PPI to customers after realising that the policies were highly profitable, and the sale of such policies



Box 2.1 Universal banking: an optimal model of banking? (continued)

was typically encouraged by large commissions. Several banks have now been fined by the Financial Conduct Authority for such practices and required to compensate customers.

A different but related issue with universal banking concerns a business model where banks are allowed to conduct their own proprietary trading and market making in addition to trading on behalf of their customers and offering other traditional banking services and products. The key concern here is that universal banks are able to leverage their cheaper sources of funds obtained from retail deposits and invest in risky projects to raise the return on equity and to meet ever-increasing shareholders' expectations. As banks expanded their risk-taking activities, both on balance sheet and on balance sheet, they became too large and complex. In addition, they exposed retail depositors to very high levels of risk-taking. This issue is particularly relevant in the US and UK, where commercial and investment banks were historically separated. It is less relevant in other European countries, where there is a long tradition of universal banking.

The global financial crisis and the subsequent international debate have prompted regulatory authorities to reflect on whether universal banking remains an optimal model of banking. Ten years on from the crisis, the answer to the universal banking debate seems to be in favour of the universal banking model. Its supporters argue that diversification can be beneficial for banks and financial stability. If one area of business underperforms, others can make up for it, making the whole business more resilient. It can also be argued that banks that are diversified in terms of businesses or geographical operations are better positioned to offer customers the depth and breadth of services these customers require, particularly as economic activities become ever more specialised and global.

In addition to economies of scale derived from size, universal banking also enables economies of scope. Broadening customer access to financial products enables increased flows of savings through the system, facilitating efficient investment and economic growth. These advantages are not to be dismissed lightly. Several European countries have a long tradition of universal banking and most banks did not turn into aggressive risk-takers, although the potential danger exists.

Regulators must ensure that banks offer their retail customers a high degree of protection. Banks must take extra care in marketing and distributing investment products to their customers.

2.4 Banking services

Modern banks offer a wide range of financial services (Figure 2.2), including:

- payment services;
- deposit and lending services;
- other financial services;
- trade finance and cross-border services.

The following sections offer an overview of payment services, deposit and lending services and other financial services. Trade finance and cross-border services will be discussed in Chapter 4.

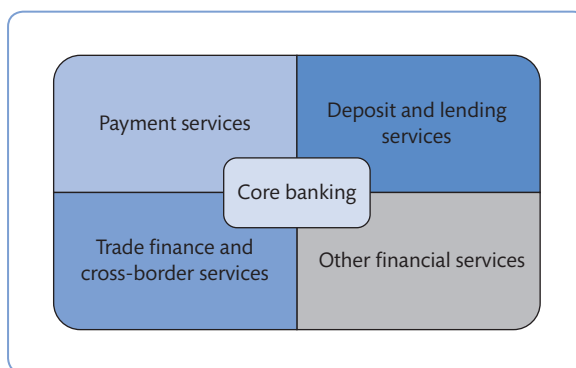


Figure 2.2 Banks' core activities

2.4.1 Payment services

An important service offered by banks is that they offer facilities that enable customers to make payments. A **payment system** can be defined as any organised arrangement for transferring value between its participants. Heffernan (2005) defines the payment systems as a by-product of the intermediation process, as it facilitates the transfer of ownership of claims in the financial sector. These payment flows reflect a variety of transactions: for goods and services as well as financial assets. Some of these transactions involve high-value transfers, typically between financial institutions. However, the highest number of transactions relates to transfers between individuals and/or companies. If any of these circulation systems failed, the functioning of large and important parts of the economy would be affected. Banks play a major role in the provision of payment services. Payments services can be either paper-based or electronic and an efficient payments system forms the basis of a well-functioning financial system. In most countries the retail payments systems are owned and run by the main banks (see also Section 3.6.1.1 on payment services).

For personal customers the main types of payments are made by writing cheques from their current accounts (known as 'checking accounts' in the US) or via debit or credit card payments. In addition, various other payment services are provided including giro (or credit transfers) and automated payments such as direct debits and standing orders. Payments services can be either paper-based or electronic and an efficient payments system forms the basis of a well-functioning financial system. In most countries the retail payments systems are owned and run by the main banks. Note that the importance of different types of cashless payments varies from country to country.

2.4.1.1 Cheques

Despite the rise of the 'cashless society' (see Box 2.2), **cheques** are still widely used as a means of payment for goods and services. If individual A buys goods and gives a cheque to individual B, it is up to B to pay the cheque into their own bank account. Individual B's bank then initiates the request to debit individual A's account. Individual A's bank authorises (clears) the cheque and a transfer of assets (settlement) then takes place. Cheque payments are known as debit transfers because they are written requests to debit the payee's account. Although not all accounts come with a cheque book, the vast majority do and in some countries (such as the UK) they are provided for free.

Nowadays cheques are primarily used to pay small businesses and bills although in this latter case discounts are frequently available if alternative methods are chosen, such as the direct debit. Cheques are often refused in shops, because of the risks that they may bounce back if not covered with enough liquidity in the payees' current account. Shops may ask for a cheque guarantee card that guarantees the cheque up to a certain amount; however the relative importance of these cards is fading and, for example, in the UK in June 2011 they were phased out. This move was part of a broader plan of the UK Payments Council (the organisation that sets strategy for UK payments) to close the central cheque clearing by the target date of 2018. This plan provoked reactions from various groups that would have been affected by the move, such as consumers' groups, organisations representing the elderly, charities (who receive a large proportion of their donations by cheque) and other bodies. After a Treasury enquiry in July 2011 the Payment Council abandoned the plan to phase out cheques and announced that cheques will continue for 'as long as customers need them'. Nonetheless, cheque usage has been steadily declining, both in value and in volume, as illustrated in Figure 2.3.

2.4.1.2 Cashless Payments

The choice of how to pay is nowadays wider than ever. In general terms, we can define a **cashless payment** as a payment made without need of physical money. Typically these payments are made by transferring money electronically, either via bank transfer or payment cards. In addition, recent developments include mobile wallets and QR codes (see Figure 2.4). The payment landscape is increasingly complex, due to the entry of new market participants and new technologies. New entrants include the large tech companies, such as Google, Amazon, Apple and Facebook, as well as smaller and innovative providers. Emerging technologies include artificial intelligence (AI), distributed ledger technology (DLT) and cloud computing. In what follows, we will briefly discuss the most commonly used methods of payment.

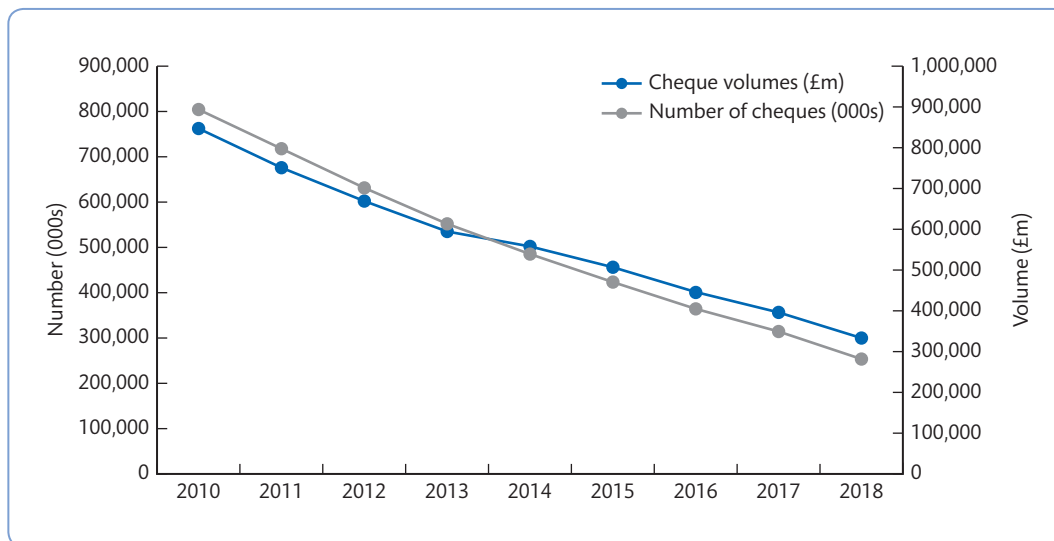


Figure 2.3 Cheque transactions in the UK

Source: Authors' elaborations, data from www.ukfinance.org.uk

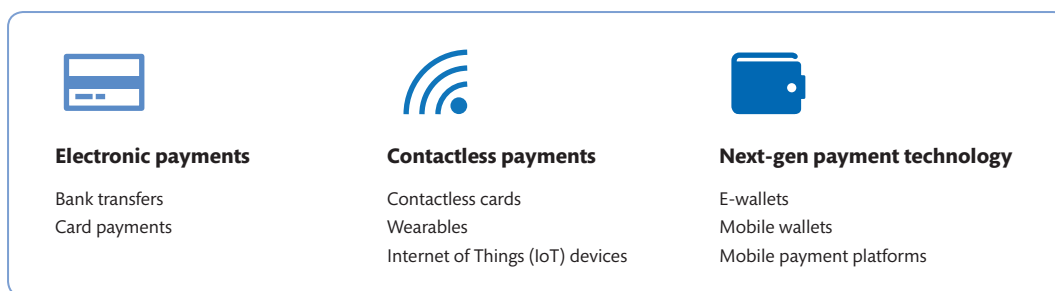


Figure 2.4 The way we pay

Cashless payments via bank transfer

- **Credit transfers** are payments where the customer instructs their bank to transfer funds directly to the beneficiary's bank account. Consumers use bank transfer payments to pay invoices or to send payment in advance for products ordered.
- **Standing orders** are instructions from the customer (account holder) to the bank to pay a fixed amount at regular intervals into the account of another individual or company. The bank has the responsibility for remembering to make these payments. Only the account holder can change the standing order instructions. Some banks will accept instructions by telephone or internet banking.
- **Direct debits** are originated by the supplier that supplied the goods/service and the customer has to sign the direct debit. The direct debit instructions are usually of a variable amount and the times at which debiting takes place can also be either fixed or variable (although usually fixed). If a payment is missed, the supplier can request the missed payment on a number of occasions. If the payments are continually missed over a period of time, the customer's bank will cancel the direct debit. Many retail customers pay utility bills (electricity, gas, water bills) in this way.

Card payments

Card payments include credit cards, debit cards, cheque guarantee cards, travel and entertainment cards and shop cards. Technically, **plastic cards** do not act themselves as a payment mechanism – they help to identify the customers and assist in creating either a paper or electronic payment.

- **Credit cards** provide holders with a pre-arranged credit limit to use for purchases at retail stores and other outlets. The retailer pays the credit card company a commission on every sale made via credit cards and the consumer obtains free credit if the bill is paid off before a certain date. If the bill is not fully paid off then it attracts interest. Visa and MasterCard are the two most important bank-owned credit card organisations. Credit cards have become an increasingly important source of consumer lending particularly in the UK and US. For example, in 1971 there was only one type of credit card (Barclaycard) available in the UK and by the early 2000s there were around 1,300. The use of cashless instruments has continued growing in the first two decades of the new millennium, although the use of cash remains persistent in many countries (see Box 2.2).

- *Pre-paid cards* are a form of pay-as-you-go credit card onto which you need to first deposit your money, then use it to pay for goods or services. Unlike normal credit or debit cards, you only spend the amount that you put on the card. Pay as you go credit cards are becoming increasingly popular for several reasons. The requirements for approval are very basic (in some cases it is as simple as being a UK resident) and are therefore available to people with poor credit history or to those who cannot legally obtain credit cards (for example, teenagers). They are also used by people who wish to be more in control of their finances, for example using as specified, pre-charged amount, as holiday money.
- *Pre-paid non-bank cards* are used to access certain services (for example, telephony services, including voice and data). Other examples are gift cards, a convenient way to offer money without delivering cash to the beneficiary. Popular gift cards include Amazon cards and App Store cards, among others.
- *Travel and entertainment cards (or charge cards)* provide payment facilities and allow repayment to be deferred until the end of the month, but they do not provide interest-free credit. Unlike credit cards, all bills have to be repaid at the end of the month and no rollover is allowed. Typically, unpaid balances are charged at a higher interest rate than for credit cards, to discourage late payment.
- *Debit cards* are issued directly by banks and allow customers to withdraw money from their accounts. They can also be used to obtain cash and other information when used through automated teller machines (ATMs).
- *Contactless payments* are increasingly available to consumers. Contactless card payments are debit or credit card payments that are processed without the need for the card to be placed in a reader or have a PIN (personal identification number) entered. Contactless payments are becoming increasingly common on a range of devices in addition to cards, including wearable devices (such as watches and wristbands) and mobile devices, such as smartphones and tablets. Contactless card payments are a faster option when paying, as there is nothing to sign and no numbers to enter. They are used for small transactions – currently up to £45 in the UK and are processed simply by holding a card or other device up to a card reader. The underlying technology for all of these devices is the same: the contactless device contains an antenna so that when it is touched against a contactless terminal, it securely transmits purchase information.
- **E-wallets** or digital wallets are electronic devices that store a user's payment details in one location. Money can be deposited in the digital wallet prior to any transactions or, in other cases, an individual's bank account can be linked to the digital wallet. Digital wallets can be used in conjunction with mobile payment systems, which allow customers to pay for purchases with their smartphones. Examples of e-wallets are Google Pay, Apple Pay and similar softwares.

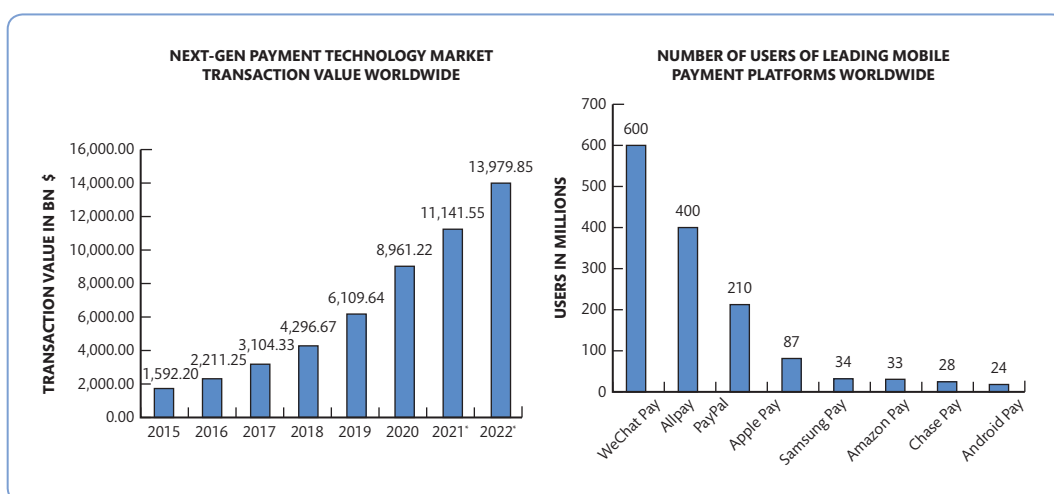
Ever since the first general-purpose charge card was introduced in the early 1950s, commentators have been predicting the 'cashless society'. However, the use of cash has proven persistent in many countries. Therefore, understanding the demand for cash and non-cash payment methods is an important policy concern. The last decade has seen a fast technological innovation and the adoption of new generation technologies, as illustrated in Figure 2.5. Nevertheless, there are still important cross-country differences and different countries are adopting different technologies. In addition, in some countries, the use of cash remains predominant, due to a combination of cultural factors, an inadequate infrastructure and security concerns. While a cashless society has many benefits, in terms of faster and more

Table 2.2 A cashless society: advantages and disadvantages

	Advantages	Disadvantages
For customers	- Cheaper, faster and more secure transactions - Easier to track payments - Helps building a credit history and improve credit scores - Low environmental impact	- Can lead to bad spending habits - Can make it harder to budget - Privacy issues - More at the risk of financial fraud - Might negatively affect those on low income or with financial difficulties
For society	- Reduction in tax avoidance - Reduction in crime - Supports economic activity and fosters growth - Supports online commerce - Fosters financial inclusion, by enabling easy money transfers	- Vulnerable to IT risks and cyber-attacks - Reduces financial inclusion as older customers, less educated customers or those with physical and cognitive disabilities might feel more confident using cash - Costs are often passed on to retailers and consumers - Can negatively impact community spirit and high street shopping - Insufficient coordination among providers, regulators and governments may lead to inefficiencies

secure transactions, it can also disadvantage certain individuals within a population, such as people with restricted or no access to financial services, older customers and customers with physical or cognitive disabilities. Table 2.2 summarises the pros and cons of a cashless society (see Box 2.2).

Figure 2.5 shows recent statistics and a forecast of the next-generation payment technology market worldwide, from 2015 to 2022. The next-generation payment technology market

**Figure 2.5** 'Next gen' payment technologies

Source: Data available at www.statista.com/statistics/788318/worldwide-payment-technology-transaction-value-market-size/

* Estimates

BOX 2.2 CORONAVIRUS ACCELERATES SHIFT AWAY FROM CASH

Andrew Crook restarted his fish and chip shop in Chorley, Lancashire, earlier this month after a coronavirus-induced lockdown, but his till remained closed. “I have taken no cash since I opened — and I love it,” he said. Mr Crook is one of many high street operators from corner shops and takeaways to launderettes and hairdressers that have long relied on hard cash to keep their businesses running. However, spurred by government guidance that recommends contactless card payments rather than cash as businesses reopen, this is changing fast. And the shift may outstay the virus. “It’s much more hygienic and easier accounting wise without,” he said, adding that card transactions had been steadily growing since he started taking them seven years ago. “I can’t see how offering cash-only transactions can be viable for any business after this.” The effects are being felt through the banking system. ATM transaction volumes fell as much as 62 per cent year on year at the start of the UK lockdown, according to data from Link, which runs the UK’s cash machine network. Volumes have picked up slightly over the past week as more businesses began to reopen, but Link chief executive John

Howells said there would not be much more of a bounce back.

“We will probably end down about 30 to 40 per cent,” Mr Howells said. He added that many reluctant consumers were “learning how to use digital alternatives” and were happy with the change. There is a similar trend in many European countries, where cash use has declined significantly during the pandemic. The daily value of cash withdrawals from ATMs fell 40 per cent after the Irish government introduced lockdown measures. The decline in the UK was not as precipitous as some countries — cash volumes in Spain dropped as much as 90 per cent — although withdrawals in the US and Russia spiked amid signs of cash hoarding. Tellingly for the future, bankers said cash volumes have not bounced back to pre-Covid levels in countries such as Austria, which has been among the first to reopen consumer-facing sectors such as bars and restaurants.

“We all knew there was going to come a point where society was virtually cashless and we’d have to work really hard to include everyone,” said Natalie Ceeney, a former civil servant and

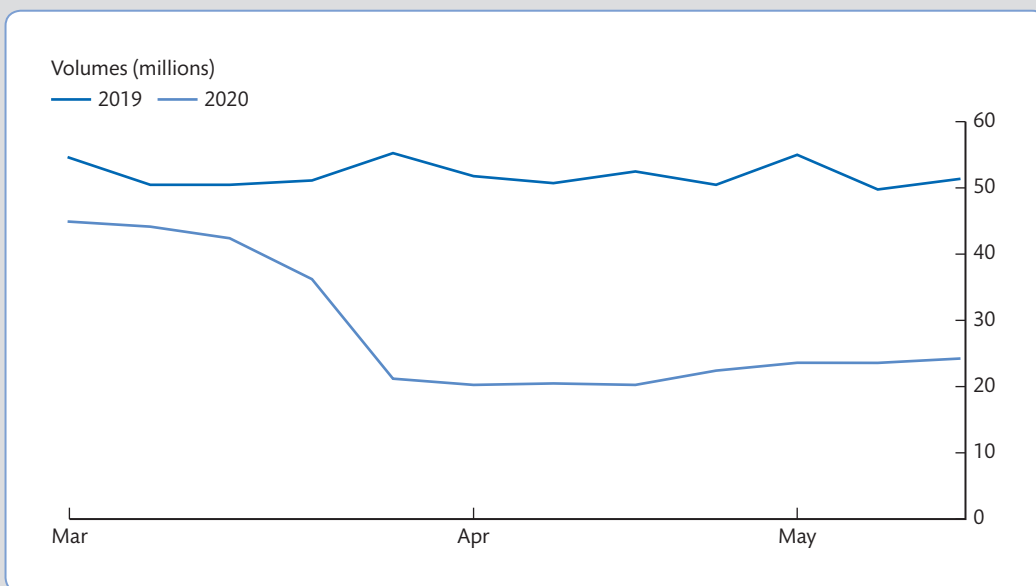


Figure 2.6 ATM transactions crash in the UK

Box 2.2 Coronavirus accelerates shift away from cash (*continued*)

banker who led an industry review into the future of cash last year. “When we finished our review a year ago we thought it would be in 10 to 15 years; Covid might have accelerated that to next year.” A decade ago, more than 60 per cent of payments were made using notes and coins. But debit cards overtook cash in 2017. Before the pandemic, bank lobby group UKFinance predicted cash would make up less than 10 per cent of payments by 2028 — now such estimates are being revised. The virus has fuelled trends that already had momentum. Notably the shift to online retail, as people stayed at home, and relied on internet shopping to get them through the crisis. The lockdown in response to coronavirus hit areas of the economy that traditionally account for much of cash payments: retail, pubs and restaurants, and travel. Even as businesses in these sectors reopen following the easing of the lockdown, bank executives predict cash will remain scarce.

The UK government has told businesses to encourage people to pay by card rather than hand over notes and coins to help prevent the spread of the virus. Some retailers said the move would be a necessity anyway: while early concerns about the risk of spreading coronavirus through handling notes has been played down, retailers report that

there is still a reluctance among customers to use cash. Some village stores have even reintroduced a ‘tick’ system, where people pay for their goods at the end of a month based on an electronic ledger — methods that have not been used for decades since shopkeepers kept large thick bound ledgers of what was owed and by who.

“The pandemic has accelerated already detailed discussions about maintaining the role of cash for those that need it,” said Stephen Jones, chief executive of UK Finance, pointing to how this would shape the infrastructure around wholesale cash distribution and cash access. Cash machines do not need to be refilled as frequently, he said. Vans that had been booked to transport cash around the country have instead been offered to help deliver resources for the pandemic response. “Banks and financial institutions have worked hard to keep cash stocks at the right level to meet demand during this period,” said Mr Jones. “It’s too early to understand what the full impact of Covid-19 and lockdown will be on the use of cash. But there will always be a role for this method of payment — it will continue to be needed.”

The shortfall in cash volumes also poses problems. Mr Howells said the sudden drop in usage could make it unsustainable to continue providing

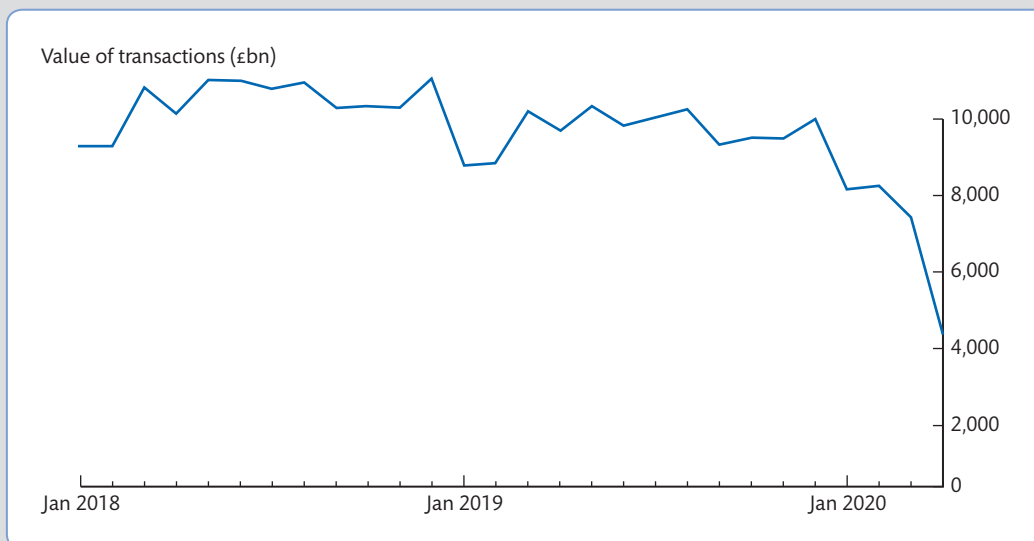


Figure 2.7 UK goes cashless during lockdown

Box 2.2 Coronavirus accelerates shift away from cash (*continued*)

access to cash for the millions of consumers who still depend on it. “When you suddenly halve the volumes, that means the business models of 40 or 50 companies can’t possibly stay the same — it’s just economically not possible,” said Mr Howells. “We are going to have changes; Link’s concern is if there isn’t some degree of co-ordination then what will happen is the quieter and more remote places will lose access.”

The UK’s cash infrastructure costs about £5bn a year to run, according to a 2019 review of the sector, with high fixed costs such as dozens of warehouses used to store and distribute coins and notes. Ms Ceeney said the government guidance over contactless payments also risked excluding some of the most vulnerable members of society. “Cash acceptance has been a minor part of the debate up to now but will increasingly be a focus, there is no reliable data in the UK on cash acceptance but anecdotally we can all see around us an awful lot of shops saying ‘contactless only.’”

Banks and ATM providers have been locked in a dispute over how to fund the cash machine network since before the pandemic hit. Marc Terry, managing director at Cardtronics, the world’s largest ATM operator, said: “It’s a bit of a self-fulfilling prophecy that we’re moving towards cashless when in fact what’s happening is we’re creating it by depriving people of the ability to make a choice.”

Customers will ultimately have their say about whether the likes of Mr Crook and the fish and chip industry can limit options on cash payments. Older people in particular will typically want to feel notes and coins in their pockets, bankers said. Mr Terry said some small businesses that initially reopened from lockdown without accepting cash were already changing course after finding some customers refuse to use cards. “As the lockdown has eased they realised they’re leaving money on the table, shops rely on every single pound that comes through the door.”



Source: Megaw, N. and Thomas, D. (2020) ‘Coronavirus accelerates shift away from cash’, *Financial Times*, 27 May. © The Financial Times Limited 2020. All Rights Reserved.

includes EMV (Europay, Mastercard and Visa) chip, QR (quick response) code, and near-field communication (NFC)/contactless payment systems. The NFC segment is currently growing at the highest rate due to its adoption in mobile handsets and wearables; however, EMV (named after its developers, Europay, MasterCard and Visa) chips are expected to dominate the market by 2022.

2.4.2 Deposits and lending services

In addition to payment services, personal banking includes the offer of a broad range of **deposit and lending services**. These are summarised as follows:

- *Current or checking accounts*, which typically pay no (or low) rates of interest and are used mainly for payments. Banks offer a broad range of current accounts tailored to various market segments and with various other services attached. As an example, Figure 2.8 illustrates how a typical current account can be used in the UK.
- *Time or savings deposits*, which involve depositing funds for a set period of time for a pre-determined or variable rate of interest. Banks offer an extensive range of such savings products, from standard fixed term and fixed deposit rate to variable term with variable rates. All banks offer deposit facilities that have features that are a combination of time and current accounts whereby customers can withdraw their funds instantly or at short

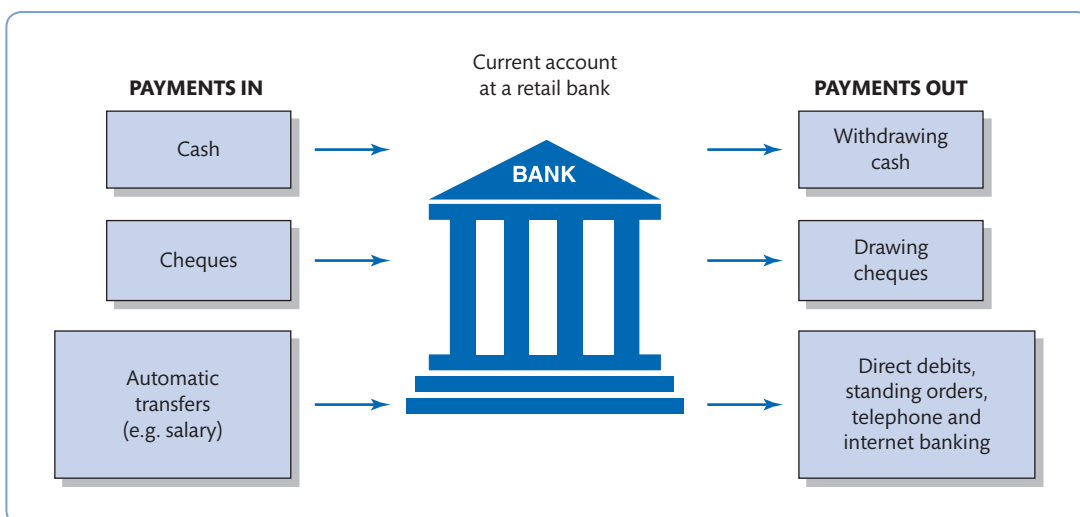


Figure 2.8 Typical current account usage

notice. Typically deposits that can be withdrawn on demand pay lower rates than those deposited in the bank for a set period.

- *Consumer loans and mortgages* are commonly offered by banks to their retail customers. Consumer loans can be unsecured (i.e. no collateral is requested; such loans are usually up to a certain amount of money and for a short to medium time period: for example, in the UK unsecured loans are up to £25,000 and repaid over five years) or secured on property (typically from £20,000 to £100,000 and repaid over ten years) and interest rates are mainly variable (but can be fixed). In addition banks of course offer an extensive array of mortgage products for the purchase of property. The main types of UK mortgages (which typically extend for 20–25 years) include: *variable rate* (interest payments vary relative to a benchmark rate such as the bank's standard lending rate or those determined by outside bodies such as the Bank of England's base rate); *fixed rate* (rates are fixed for a set period, usually 2–5 years, and then revert to variable rate); *capped* (rates vary but a cap is placed on the maximum rate paid over a specified period); *discount mortgages* (where rates vary but are discounted at a few percentages below a benchmark rate over a period – e.g. 1 per cent discount to the base rate over the first two years); and *cashback mortgages* (where those taking out the mortgage receive a single lump sum or cashback generally based on the value of the loan). Mortgages can also be obtained in foreign currency, for the purchase of overseas properties, and also for 'buy-to-let' property.

Alternative mortgage products, that is mortgages that are different from the traditional principle and interest repayment schedule, have been identified as some of the culprits of the 2007 sub-prime crisis (mainly the adjustable rate mortgages – ARM – which earned the nickname of 'toxic mortgages'). However, research by Cocco (2013) argues that because of their lower initial mortgage payments relative to loan amount, they may be a valuable tool for households that expect higher and more certain future labour income, and that wish to smooth consumption over the life-cycle. The size and shape of the mortgage market has changed considerably over the last decade since the global financial crisis, as new prudential rules designed to protect both borrowers and banks came into force. At the

same time, the UK residential mortgage market has seen significant government intervention to stimulate new home building and support first-time buyers, particularly through the Help to Buy equity loan scheme (HTBEL), which was launched in 2013 to help those with small deposits buy new homes. However, the scheme is due to end in 2023, with a potentially negative impact upon new house building and ownership.

Although the full effect of the Covid-19 health crisis on the housing market is yet to be seen, the first six months of 2020 have seen a decrease in both mortgage applications and mortgage approvals, as both lenders and borrowers are still dealing with the economic impact of the lockdown.

2.4.3 Investment, pensions and insurance services

- **Investment products** offered to retail customers include various securities-related products: mutual funds (known as unit trusts in the UK), investment in company stocks and various other securities-related products (such as savings bonds). In reality, there is a strong overlap between savings and investment products and many banks advertise these services together.
- **Pensions and insurance services** are nowadays widely offered by many banks. Pension services provide retirement income (in the form of annuities) to those contributing to pension plans. Contributions paid into the pension fund are invested in long-term investments with the individual making contributions receiving a pension on retirement. The pension services offered via banks are known as private pensions to distinguish them from public pensions offered by the state. Usually there are tax advantages associated with pensions contributions as most governments wish to encourage individuals to save for their retirement. Insurance products protect individuals (policyholders) from various adverse events. Policyholders pay regular premiums and the insurer promises compensation if the specific insured event occurs. There are two main types of insurance – life insurance and general (or property and casualty) insurance. The latter is insurance that does not involve death as the main risk. It includes home, travel, medical, auto and various other types of insurance. Banks offer both life and non-life insurance products with the latter mainly being travel, property, mortgage repayment and other types of protection. In the UK, there has also been substantial growth in income protection insurance (insurance that replaces earnings if individuals are unable to work) and critical illness insurance (which covers medical costs and/or income).
- *Payment protection insurance* (also known as PPI) is a type of insurance sold with loans, credit cards, mortgages and other types of credit. It was designed to meet the cost of repayments should customers become unemployed or ill. On some loans, the whole cost of the PPI premium was added upfront to the amount borrowed. On others, including mortgages, borrowers mostly paid for PPI by monthly instalments. Most customers were pressurised into taking out the policy, which was presented as a necessary condition to obtain credit (House of Commons, 2016). This resulted in the UK's largest consumer mis-selling scandal, which resulted in lenders collectively settling claims amounting to almost £50bn (see also Box 2.3). While the financial industry insists a scandal of the proportion of the PPI mis-selling could not happen again, regulators and consumers groups worry that the progress achieved could reverse during an economic recession. Retail banks' basic

profitability is low, and regulators fear an increased drive to generate income from new products, particularly investment products.

Where could the next PPI scandal arise? It might not be the big banks; other financial intermediaries are currently under the regulators' close watch. For example, payday lending (a type of short-term high-interest borrowing) grew substantially since the global financial crisis, as large banks became increasingly reluctant to lend to high-risk borrowers. In 2018, Wonga, the UK largest payday lender, collapsed as it was inundated with complaints from former customers. Another sector that might be susceptible to mis-selling is the insurance sector, particularly pension policies. UK regulators fear a major mis-selling scandal as customers who have transferred out of final-salary pension schemes might have received poor advice.

BOX 2.3 AFTER PPI, WHAT COULD BE THE NEXT BANKING MIS-SELLING SCANDAL?

When Matthew started working for one of the UK's biggest banks in the late 1990s, he didn't think there was anything wrong with selling payment protection insurance. Soon, the pressure to sell policies to people who didn't need, want or understand the cover they were signing up for made him feel ill. "There were often times I would go down the back lane behind the branch and be physically sick because I was so frightened about going into work," says Matthew, who asked the

FT not to use his real name as he still works with Lloyds Banking Group. The short-term result was a cut-throat sales culture that saw stories like Matthew's repeated in thousands of bank branches across the country. The most successful account managers were rewarded with bonuses, booze and all expenses paid holidays. The long-term result was the UK's largest consumer mis-selling scandal. In those two decades, Lloyds has put aside £20bn to settle potential mis-selling

£36bn has been paid in PPI compensation since 2011

PPI payouts (£bn)

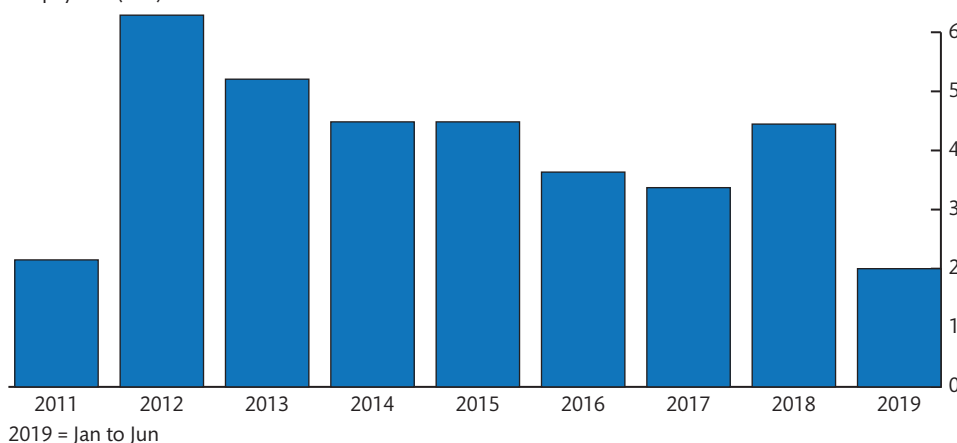


Figure 2.9 PPI payouts

Source: FCA

Box 2.3 After PPI, what could be the next banking mis-selling scandal? (*continued*)

claims, topping a list of UK lenders who collectively expect the PPI scandal to cost them close to £50bn. With the passing of a final deadline for compensation claims on Thursday evening, the finance world hopes to draw a line under the scandal. Bankers are privately keen to say they have learnt their lessons and cleaned up their act. Yet this week, one of the UK's most senior financial regulators shared concerns that banks needed "a generation" to fix the cultural issues that led to such rampant mis-selling.

Jonathan Davidson, director of supervision for retail and authorisations at the Financial Conduct Authority, told the FT that while lenders had made genuine efforts to reform, he was concerned that progress could be reversed if a recession increased pressure on their main sources of income. Today's competitive banking landscape has parallels with the 1990s and early 2000s when prices for many basic products and services were falling to unsustainably low levels. Unwilling to lose market share by raising prices, lenders instead pushed staff to cross-sell more profitable products like PPI at any cost. Some insist a problem on the scale of PPI could never happen again. Nevertheless, regulators, consumer groups, and the claims management companies that sprang up in the wake of PPI are already eyeing several new areas that could prove to be the next consumer mis-selling scandal.

Payday loans

PPI set a precedent — much maligned by executives and investors — that firms could be retroactively punished for historic practices judged to be unfair, even if they weren't against any rules at the time. Rob James, a portfolio manager and financial services analyst at Merian Global Investors, says: "It is a bit of a known unknown — we know that things could happen but we don't know when or where. My worry spreads a little wider than the banking system now." That precedent of retroactive punishment is already wreaking havoc on the UK's high-cost short-term lending sector, which has provided a lucrative — but temporary — alternative source of income for claims management companies as PPI compensation tails off. Payday lending grew rapidly in the aftermath of the financial crisis as mainstream banks

became reluctant to lend to riskier borrowers, but firms such as Wonga were criticised for extending loans with interest rates as high as 5,800 per cent. The FCA introduced a cap on charges in 2015. At the time, regulators assumed the cap would force some particularly exploitative or inefficient firms out of business, but expected more mature lenders to be able to ride out the disruption. However, the sector has since been inundated with complaints about unaffordable loans issued under the previous regulatory regime.

The Financial Ombudsman Service has been sympathetic to historic complaints, prompting a 130 per cent increase in the past financial year. Furthermore, the FOS charges companies a £550 processing fee for every complaint it handles beyond the first 25 each year, regardless of the outcome. The cost of dealing with the influx of claims from CMCs has been a major factor in the collapse of Wonga and others — showing the huge drain of their consumer redress operations. Revenues in the financial services claims management sector increased almost sixfold between 2009/10 and 2017/18, from £104m to £600m. Now the PPI deadline has passed, claims managers are seeking a new target. Several hundred claims management firms closed ahead of the PPI deadline, discouraged by more stringent regulations that came into force this year, but there are still 750 companies seeking authorisation from the FCA.

Banking and investment

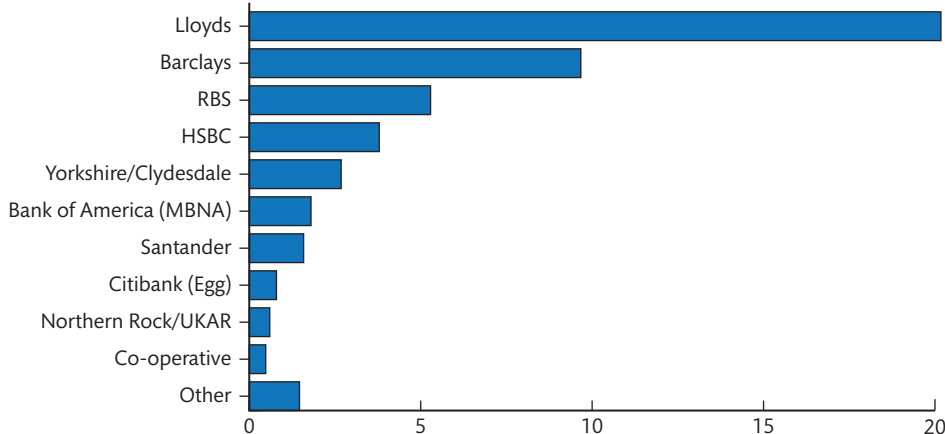
They may no longer be sending endless text messages about PPI compensation, but the claims managers are still seeking to profit from mis-sold financial products, including investments. Their aggressive tactics have provoked anger from many within the banking industry. John McFarlane, former chairperson of Barclays, was criticised last year for claiming that the companies had "turned portions of Britain into fraudsters" by encouraging spurious claims. He says his earlier comments had been "sensationalised", but maintained that the PPI process had encouraged too many illegitimate claims. "There is no doubt that banks were at fault for the way they exploited this additional source of income and for not putting in place adequate procedures to ensure people weren't exploited," Mr McFarlane says,



Box 2.3 After PPI, what could be the next banking mis-selling scandal? (continued)

PPI has cost UK banks £48bn

Total cost of PPI per bank, including compensation and administration, £bn

**Figure 2.10** Total cost of PPI per bank, including compensation and administration (£bn)

Source: New City Agenda

but “we’ve moved on now”. Consumers may be sceptical of such promises, but the threat posed by claims managers means financial companies now face a major economic deterrent against bad behaviour. The history of PPI is littered with opportunities for banks to have limited the eventual size of the scandal. The first legal judgment against mis-selling came as early as 1993, but the case was kept from the public under a 10-year confidentiality clause that allowed lenders to keep on profiting. When complaints picked up pace after 2005, the banks resisted at every stage until they were defeated in a high court battle in 2011. Natalie Ceeney, former head of the Financial Ombudsman Service, was involved in this battle. “Crudely, the economics were that a bank could do a lot of harm — and make a lot of profit — but only a tiny percentage of customers would complain,” she says. To satisfy those customers, “all you’d need to do [was] give them their money back. But that’s all changed. If something happens now, there is a huge claims industry ready to pounce and a regulator willing to fine. Banks can’t take that risk any more.” Ms Ceeney says the change in approach can be seen in the way banks responded to more recent criticism over products

such as packaged current accounts. Complaints from customers who were sold needless add-ons such as travel insurance alongside their current accounts began to spike in 2015, but rather than resist as they did with PPI, banks were quick to clamp down before it could become a major problem. A partner at one financial litigation specialist acknowledges that packaged accounts had not proven as lucrative for complainants as initially hoped, but says claims managers will continue to target banks in several other areas.

“The problem with retail banking is they don’t make enough cash from the basics, so they will always feel pressured to come up with new products,” the partner says. Radio adverts that previously exhorted listeners to check if they had PPI are increasingly focusing on mis-sold investment products and Isas. Customers could be eligible for compensation if they lost money on an investment if their bank or financial adviser did not properly explain the risks. Recent investment scandals, such as the £236m collapse of London Capital & Finance, have also raised important regulatory questions about how well retail investors understand the risks of products such as mini-bonds

Box 2.3 After PPI, what could be the next banking mis-selling scandal? (continued)

Table 2.3 Top ten financial scandals in UK retail banking

Type of misconduct	Provisions by UK banks (£bn)	Years provisions were incurred
PPI mis-selling	48.5	2010-19
Interest rate hedging products mis-selling	4.9	2012-15
Endowment mortgages mis-selling	1.9	2002-06
Mortgages	1.6	2002-19
Packaged bank account mis-selling	1.6	2014-18
Consumer Credit Act breaches	1.0	2013-15
Investment products and advice mis-selling	0.9	2003-15
Pensions mis-selling	0.6	2000-02
Unfair unauthorised overdraft charges	0.6	2006-07
ID theft and card protection insurance mis-selling	0.5	2014-15
Other issues / miscellaneous	5.4	2000-16
Total	67.4	2000-17

Source: New City Agenda

and Innovative Finance Isas. Many LCF investors wrongly assumed their mini-bond investments carried some degree of regulatory protection. While LCF was regulated by the FCA, mini-bonds are not, meaning they are not typically covered by the Financial Services Compensation Scheme (FSCS) unless people have invested through official advice. However, correspondence previously seen by the FT showed that even the FSCS did not spell out this nuance correctly to some bondholders, who are trying to claim compensation after being told they were protected by the guarantee scheme to the tune of £50,000. The FCA has also been highly critical of the ways Innovative Finance Isas, backed by peer-to-peer investments, have been marketed to those with cash Isas. It is now considering placing restrictions on which products unadvised retail investors can buy over the internet. The litigation partner says he also has concerns that many banking customers had paid high fees for the promise of actively managed stocks and shares Isas which carried on charging high fees when they moved to more passive investment strategies. “I think a lot of the future claims will have different characteristics to PPI,” he adds. “With products like investments,

the volumes will be less, but the quantum of each claim could be much higher.”

Pensions and business banking

Some of the most valuable mis-selling claims of all are likely to come from the largest investments — namely, pensions. The pension freedoms of 2015 were designed to give those approaching retirement more choice, but the changes have opened up a number of mis-selling opportunities. “If you have a really good defined benefit pension, it’s incredibly valuable. Consumers often don’t understand just how valuable it is to have guaranteed income for the rest of their life,” says the FCA’s Mr Davidson. In July, the FCA warned that much pension advice was “not of an acceptable standard”, and outlined plans to ban a type of “contingent charging” that incentivises advisers to encourage as many customers as possible to transfer their pensions, and then charge them a high premium for doing so. The regulator is also in the early stages of probing banks’ treatment of small business customers. Some companies are already fighting high-profile complaints from individual customers; CYBG, Lloyds and Danske Bank were all targeted by hunger strikers this



Box 2.3 After PPI, what could be the next banking mis-selling scandal? (continued)

week complaining about issues including mis-sold loans. However, a probe by the FCA could raise more fundamental questions about the area of small business lending. The regulator has so far committed only to “exploratory work” on the sector, but has raised concerns about trends such as high transaction charges on business current accounts and other services such as foreign exchange. Ranked in order of size, the mis-selling of interest rate swaps to business banking customers is the second-biggest scandal after PPI, albeit a fraction of the size at just under £5bn. Other scandals in business banking to have emerged since the financial crisis include a major fraud at HBOS’s Reading branch, and the exploitation of struggling businesses placed in Royal Bank of Scotland’s disgraced Global Restructuring Group.

A new kind of mis-selling: customer data

While the FCA’s Mr Davidson mentions both SMEs and pensions as key areas of concern, he predicts the biggest problem could come from a new type of mis-selling. “The next big question is about data. It is incredibly valuable, and it is our intent that customers benefit from the value,” he says. Established banks and technology specialists from start-ups to giants such as Amazon have spoken about their desire to use advanced data analysis to create new, more personalised types of financial products at lower costs. But a recent report from the Bank for International Settlements — the central bank for central banks — warned that firms could use the same information to exploit customers. For example, studying customer data could allow lenders to overcharge by working out the maximum rate a borrower would be willing to pay for

a loan, rather than competing to give them the cheapest available deal. Data-driven banking is in its early stages, but regulators are not alone in keeping a close eye on developments. The litigation partner adds: “This is very much a space that we are keeping an eye on.” He says that previous breaches of data rules had generally been dealt with through firm-level fines rather than by compensating individual customers, but predicts that in time “this will probably change”. The idea that the regulator could retroactively “move the goalposts” to allow customers to complain about old products was the biggest source of criticism from banks during the PPI scandal. A former executive who worked at several banks that sold large volumes of PPI argues: “At that time, when customers bought something they were thought to be responsible for knowing what they were doing. That responsibility has dramatically shifted now, but by the standards of the time, no one was deliberately ripping people off.” The FCA, however, is unmoved by such criticism. AI-influenced products and services are a new area, meaning that many boundaries on acceptable practice have yet to be agreed, and questions over how banks can use customer data are part of a broader debate on the ethics of new technology. But Mr Davidson warns companies against looking to take advantage of this ambiguity and make a quick buck before formal rules are agreed. “If your purpose or your intent is to use that data to give [customers] a better deal, then you’ll hear nothing but praise for your innovation from me,” he says. “If you use that data to make more money out of [them], then it’s not going to be a healthy relationship. And you might not be breaching some rules [right now], but that will only be [the case] for a while.”



Source: Megaw, N. (2019) ‘After PPI, what could be the next banking mis-selling scandal?’, *Financial Times*, 30 August.
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2.5 Banks and the FinTech revolution

Financial technology (**FinTech**) has been one of the leading forces of changes affecting the financial industry in the past decade, and it holds great promise for the provision of financial services, from lower transaction costs to increased access to financial services.

Nevertheless, FinTech might affect market structure, competition and financial stability. For these reasons, technology-driven innovation in financial services is attracting increased supervisory scrutiny (BIS, 2018). The challenge for policy makers is to maximise the benefits of FinTech while minimising potential risks for the financial system (BIS, 2020). Achieving this balance is crucial to ensure that innovation can benefit society and foster economic growth without compromising systemic stability.

Generally speaking, FinTech can be defined as the use of innovative technology in financial services. The Financial Stability Board (FSB) defines FinTech as *'technologically enabled financial innovation that could result in new business models, applications, processes or products with an associated material effect on financial markets and institutions and the provision of financial services'*.¹ This is a broad definition, but it is useful in that it takes into account the fast-paced changes in the sector. The key drivers of change of FinTech innovations are technology, regulation and evolving customer preferences.

Several factors affect both the supply and the demand side of financial services (FSB, 2019). On the supply side, technology and regulatory changes are the key drivers of changes. On the demand side, a generation of digital natives is changing consumer behaviours and expectations at a faster than ever pace.

The Bank for international settlements (BIS, 2020a) offers an attempt to classify these innovations. Building on the works of global standard-setting bodies, the BIS proposed a conceptual framework to analyse policy responses to FinTech, referred to as the FinTech tree (see Figure 2.11). The FinTech tree distinguishes three categories: FinTech activities, enabling technologies and policy enablers. FinTech activities (for example, **digital banking** or robo-advice) can take various forms and may be performed in different sectors of the financial industry. Enabling technologies (for example, cloud computing or artificial intelligence) are those that make innovation possible in the provision of financial services and, as such, form the backbone of FinTech activities. Policy enablers refer to public policy measures and initiatives (e.g. digital ID systems) that support the development of FinTech activities and the use of enabling technologies.

The main innovations in different market sectors are summarised in Figure 2.12. The highest number of new FinTech service providers are in the payment space, followed by credit, deposits and capital-raising services.

Technology is also helping the unbundling of financial services, and the use of data will become increasingly important. Up to now, FinTech firms have specialised in specific markets, and their relationship with banks has been mostly complementary. FinTech start-ups have so far lacked the capital and the scale to compete with large financial institutions. They have therefore tended to co-operate in niche market segments (for example, in the peer-to-peer lending platform space). This dynamic might change with the entry of larger and established technology companies, the so-called Big Tech (Google, Apple, Amazon, Facebook, but also Alibaba, Baidu, and Tencent). These companies can leverage their access to client data to acquire market share and are becoming increasingly active in several areas, particularly in mobile payments. For example, AliPay is the largest mobile payment platform in China, and M-Pesa in Kenya provides a considerable part of local banking services.

¹ See www.fsb.org/what-we-do/policy-development/additional-policy-areas/monitoring-of-fintech/

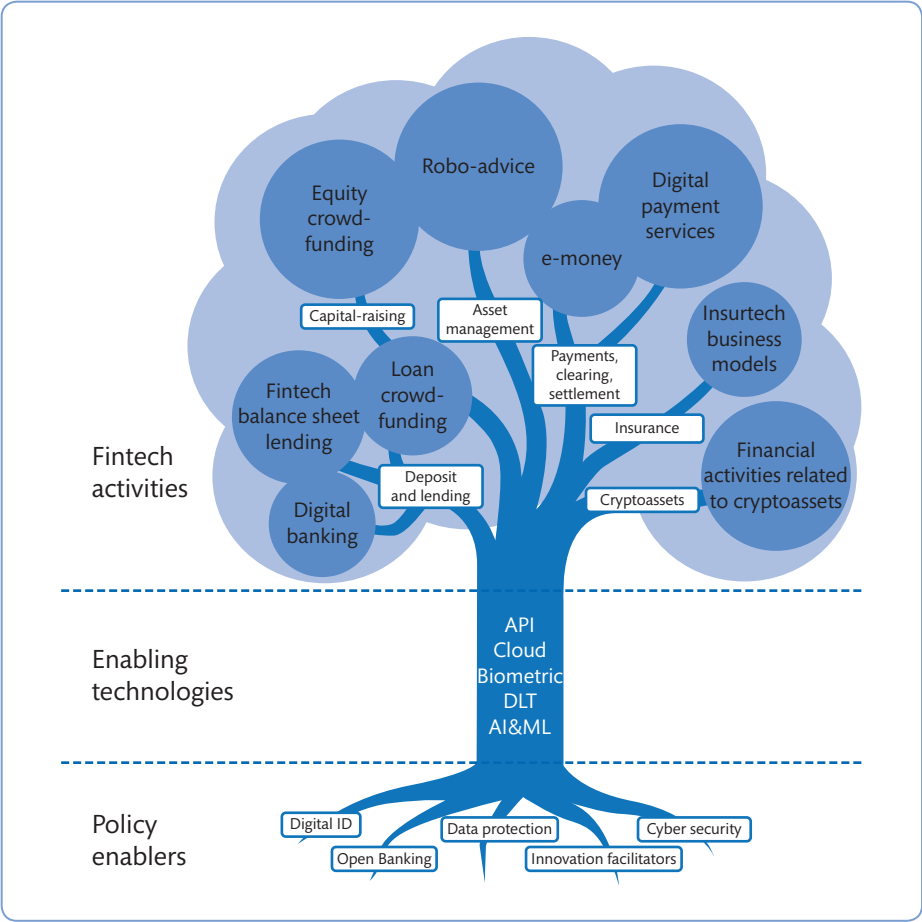


Figure 2.11 The FinTech tree

Source: Adapted from Bank for International Settlements (2020a).

Credit, deposit and capital raising services	Payments, clearing and settlement services	Asset management services	Insurance	Crypto assets
• Crowdfunding • Equity crowdfunding • Lending marketplaces • Mobile banks • Digital banks • Credit scoring	• RETAIL • Mobile wallets • P2P transfers • Digital currencies • E-money services • WHOLESALE • Value transfer networks • FX wholesale • Digital exchange platforms	• High frequency trading • Copy trading • E-trading • Robo advice	• Insurtech business models refer to technology-driven innovative models emerging in: (i) distribution (comparison portals and digital brokers); and (ii) underwriting (mobile, on demand, usage-based or technology-enabled P2P and parametric insurance)	• A type of private assets that depends on cryptography and DLT technology

Figure 2.12 FinTech innovations by sector

Source: Adapted from Bank for International Settlements (2020a).

2.5.1 What will be the impact of the FinTech revolution on the banking industry?

A recent Basel Committee on Banking Supervision survey considered five different scenarios, ranging from a better banking industry, where incumbent institutions overcome their legacy infrastructure to disintermediation, with FinTech and Big Tech offering all financial services (BCBS, 2018a). We'll revise these five scenarios below.

- *Better bank* – under this scenario, incumbent banks modernise their infrastructure, digitise their provision, and retain customer relationships. Banks continue offering core financial services and adopt new technologies (from AI to DLT) to change their business models and achieve cost reductions. There are early signs that incumbent banks are investing heavily in technology, and some are increasingly innovative. There are, however, legacy issues for many banks and whether most institutions will manage to leverage technology to remain competitive remains to be seen.
- *New bank (or neo bank)* – the emergence of a new type of digital bank (also known as challenger banks) may point to a scenario whereby incumbents do not survive the technological disruption, and neo banks, built on digital platforms, emerge as key players. Examples of new banks include Monzo and Revolut in the UK, N26 in Germany, WeBank in China and similar institutions in many countries. While we have seen a large number of neo banks emerging in recent years, many still lack the scale to pose a serious threat to incumbents.
- *Distributed bank* – under this scenario, financial services become increasingly specialised and modularised, with a large number of new businesses emerging to provide 'plug and play' services on digital interfaces. In this scenario, banks and FinTechs compete for customers but also co-operate, in niche markets. Elements of this scenario are currently playing out, for example, in the payment space. Innovative payment services are emerging as a result of joint ventures, fostered by the use of APIs. For example, Apple Pay, Samsung Pay and Android Pay are examples of this distributed scenario as Bigtech payments solutions layer existing card networks.
- *Relegated bank* – in this scenario, incumbent banks will leverage their banking licence to provide deposit and lending services and other regulated financial services, but tech companies manage the customer relationship. In this scenario, cloud computing and AI are fully exploited by front-end platforms that make innovative and extensive use of connectivity and data to improve the customer experience. Banks are relegated to being providers of services to the platforms, and the latter manage customer relationships. Although this scenario might appear unlikely, there are already examples of modularised financial services, particularly in the payment industry. Other examples include online lending platforms. Social media platforms can leverage customer data to offer its customers tailored financial products and services from third parties, including banks. An example is the instant messaging application WeChat in China.
- *Disintermediated bank* – in this final scenario, banks are no longer significant players because the need for balance sheet intermediation or a trusted third party is removed. Banks are displaced from customer financial transactions by more agile platforms and technologies, which ensure a direct matching of final consumers depending on their financial needs. Some elements of this scenario can be seen, for example, in P2P lending platforms and cryptocurrencies (for example, Bitcoin).

In sum, the future of the banking industry may depend on incumbent banks' ability to adapt and innovate. It is also important to note that while the emergence of new technologies offers many benefits to consumers, there are also potential risks, from cyber risk to data security, that might warrant an expansion of the regulatory perimeter. Thakor (2020) provides a good discussion of these issues, together with a review of the academic literature in this area.

2.6 Conclusions

The nature of the banking business has changed dramatically. From traditional lending institutions, thanks to deregulation nowadays banks can offer a wide variety of financial services and products. This means that they now compete with non-bank financial institutions and operate as universal banks offering non-banking products such as insurance and asset management. Similarly, there has been a considerable evolution in financial technology (FinTech), which promises to revolutionise the banking sector. In recent years we have seen important changes in the payment systems, mainly as a result of technological advancements and innovations and a change in demand of consumers. Not surprisingly, cheques usage has dropped sharply in most countries while the relative importance of mobile banking services has increased. In this chapter, we have discussed banks' core activities and the key products and services they offer, with a focus on the impact of technological innovation. We have reviewed different future scenarios, ranging from a positive one where banks will leverage the technological advances to innovate and offer better customer experiences, to a negative one where banks become irrelevant as the need for balance sheet intermediation is removed. The likelihood of each of these scenarios will depend on the interplay of many factors, including banks themselves, but also regulators, customers and key players in other industries.

Key terms

Cashless payments	Deposit-taking institution	FinTech	Pension services
Cheques	Digital banking	Insurance services	Plastic cards
Credit transfers	Direct debit	Investment products	Standing order
Deposit and lending services	E-money	Non-deposit-taking institution	Universal bank
	E-wallet/mobile wallet	Payment system	

Key readings

- Basel Committee on Banking Supervision (2018) 'Sound practices: Implications of FinTech developments for banks and bank supervisors', February.
- Financial Stability Board (2019), 'FinTech and market structure in financial services: Market developments and potential financial stability implications', February.
- Thakor, A. (2020) 'FinTech and banking: What do we know', *Journal of Financial Intermediation*, 41, in press.

REVISION QUESTIONS AND PROBLEMS

- 2.1** What is a deposit-taking institution?
- 2.2** Why is the ability to collect retail deposits such an important feature?
- 2.3** Define universal banking. Discuss the advantages and disadvantages of a universal banking system. Has the 2007–2009 financial crisis exposed the weaknesses of the universal bank business model?
- 2.4** Discuss the main services offered by banks.
- 2.5** How has the use of technology changed the retail banking industry?
- 2.6** Discuss the key innovations in retail payments and how these have changed the way we pay.
- 2.7** Discuss the pros and cons of a cashless society.
- 2.8** What will be the impact of FinTech on the future of the banking industry?
- 2.9** P2P lending platforms clearly cut out the ‘middle man’ by effectively disintermediating the banks. Discuss the following questions:
 - (a) What are the benefits and risks of the P2P lending process?
 - (b) In what way(s) is it different from traditional banking?
 - (c) Could this system potentially produce a better outcome for all parties involved?
- 2.10** Should FinTech activities be regulated?

Appendix 2.1

FinTech: a selected glossary

Term	Explanation
Application programming interface (API)	A set of rules and specifications followed by software programmes to communicate with each other, and an interface between different software programmes that facilitates their interaction.
Artificial intelligence (AI)	IT systems that perform functions requiring human capabilities. AI can ask questions, discover and test hypotheses, and make decisions automatically based on advanced analytics operating on extensive data sets. Machine learning (see below) is one subcategory of AI.
Big data	Designates the large volume of data that can be generated, analysed and increasingly used by digital tools and information systems. This capability is driven by the increased availability of structured data, the ability to process unstructured data, increased data storage capabilities and advances in computing power.
Bigtech	Large, globally active technology firms with a relative advantage in digital technology. The GAFA acronym refers specifically to a set of the largest technology companies, namely Google, Amazon, Facebook and Apple (the GAFAA acronym is also used to include Alibaba, the largest Chinese technology company).
Cloud computing	The use of an online network ('cloud') of hosting processors to increase the scale and flexibility of computing capacity. This model enables convenient on-demand network access to a shared pool of configurable computing resources (e.g. networks, servers, storage facilities, applications and services) that can be rapidly released with minimal management effort or service provider interaction.
Crowdfunding	The practice of funding a project or venture by raising monetary contributions from a large number of people. It is often performed today via internet-mediated registries that facilitate money collection for the borrower (lending) or issuer (equity).
Cryptoassets	A type of private asset that depends primarily on cryptography and distributed ledger or similar technology as part of its perceived or inherent value.
Cyber-crime	When a computer system or component is the object of the crime (hacking, phishing, spamming) or is the facilitator of a crime (such as theft of information or money).
Cyber resilience	A financial market infrastructure's ability to anticipate, withstand, contain and rapidly recover from a cyber attack.
Cyber-risk	The combination of the probability of an event occurring within the realm of an organisation's information assets, computer and communication resources and the consequences of that event for the given organisation.
Digital banks	Deposit-taking institutions that are members of a deposit insurance scheme and deliver banking services primarily through electronic channels instead of physical branches.
Digital ID system	A system that covers the process of identity proofing/enrolment and authentication. Identity proofing and enrolment can be either digital or physical (documentary), or a combination, but binding, credentialing, authentication, and portability/federation must be digital.
Digital payment services	Digital payment service providers make use of technology to facilitate payment transactions by transferring money, clearing or settling balances digitally, without the use of physical money.
Digital currency	An asset that only exists electronically and that can be used as a currency (means of payment, store of value, unit of account) although it is not legal tender. Digital currencies are often underpinned by distributed ledger technology (DLT) to record and verify transactions made using the digital currency. These can include private currencies and digital versions of national bank currencies. Because of the use of cryptography techniques, a (large) subset of digital currencies are referred to as 'cryptocurrencies'.

Appendix 2.1 continued

Term	Explanation
Distributed ledger technology (DLT)	Technologies, such as blockchain, that are a means of recording information through a distributed ledger, i.e. a repeated digital copy of data at multiple locations. These technologies enable nodes in a network to securely propose, validate and record state changes (or updates) to a synchronised ledger that is distributed across the network's nodes.
E-money	Money issued or stored electronically, for example, on a device such as a chip card (e.g. payment cards), a hard drive in a personal computer or a server – with the purpose of replacing physical money in payment transactions.
E-money services	Issuance of debt-like instruments for the purpose of facilitating payment transactions.
Equity crowdfunding	An activity where investors provide funding to private companies in the form of equity. The FinTech platform matches investors with companies they want to invest in, enabling them to participate in the early capital-raising activities of startups and other companies.
FinTech	Technologically enabled financial innovation that could result in new business models, applications, processes or products with an associated material effect on financial markets and institutions and the provision of financial services.
Innovation accelerator	A partnership arrangement between FinTech providers and central banks/supervisory agencies to develop cases that may involve funding support and/or authorities' endorsement/ approval for future use in central banking operations or in the conduct of supervisory tasks.
Insurtech business models	Technology-driven innovative business models that are emerging in two major areas of insurance: (i) distribution, such as comparison portals and digital brokers; and (ii) underwriting, such as mobile, on-demand, usage-based or technology-enabled peer-to-peer and parametric insurance.
Internet of Things (IoT)	The networking of physical devices, vehicles, buildings, and other items embedded with electronics, software, sensors, actuators, and network connectivity that enable these objects to (a) collect and exchange data and (b) send, receive and execute commands.
Machine learning (ML)	A method of designing problem-solving rules that improve automatically through experience. Machine-learning algorithms give computers the ability to learn without specifying all the knowledge a computer would need to perform the desired task, as well as study and build algorithms that can learn from and make predictions based on data and experience.
Mobile wallets	Replicate a physical wallet in a digital interface on a mobile phone. Customers can add credit and debit cards, as well as prepaid cards, gift cards and rewards cards to be stored and carried. This use case not only replaces physical plastic cards, but also allows those cards to be enhanced by additional services.
Neo-banks	New entrants that make extensive use of technology (e.g. API, big data, artificial intelligence, blockchain) in order to offer retail banking services predominantly through smartphone apps and internet-based platforms. Neo-banks target individuals, entrepreneurs and small to medium-sized enterprises.
Online lending platforms	Platforms that intermediate loans online, and may be operated by banks or non-banks. Some online lenders keep all or some of the loans they originate, while others sell or securitise them. Funding for loans may come from traditional sources, such as deposits, if the lender is a bank, securitisations, private investors and capital raisings, and loans from banks. Additionally, funding may come from 'peer-to-peer' arrangements that directly match lenders with borrowers via online platforms. Together with non-loan forms of finance such as invoice trading, these models make up the 'FinTech credit' category.
Open banking	The sharing and leveraging of customer-permissioned data from banks with third-party developers and firms to build applications and services that provide real-time payments, greater financial transparency options for account holders, marketing, etc.
Regtech (regulatory technology)	Any range of FinTech applications for regulatory reporting and compliance purposes by regulated financial institutions. This can also refer to firms that offer such applications.

Appendix 2.1 continued

Term	Explanation
Regulatory sandbox	A controlled testing environment, sometimes featuring regulatory forbearance and alleviation through the use of legally provided discretions by the supervisory agency. The testing environment may involve limits or parameters within which the firms must operate (e.g. restrictions on the time a firm may operate in the sandbox).
Robo-advisors	Applications that combine digital interfaces and algorithms, and can also include machine learning, in order to provide services ranging from automated financial recommendations to contract brokering to portfolio management to their clients, with limited human intervention or none. Such advisors may be standalone firms and platforms, or can be the in-house applications of incumbent financial institutions.
Security biometric services	A security mechanism used to identify, authenticate and provide access to a facility or system based on the automatic and instant verification of an individual's physical characteristics, such as fingerprints, retina patterns etc.
Smart contracts	Programmable applications that, in financial transactions, can trigger financial flows or changes of ownership if specific events occur. Some smart contracts are able to self-verify their own conditions and self-execute by releasing payments and/or carrying out others' instructions.
SupTech (supervisory technology)	The use of technologically enabled innovation by supervisory authorities.

Source: Adapted from Basel Committee on Banking Supervision (2018) and Ehrentraud *et al.* (2020).